

Multiple *wh*-sluicing in Korean: An experiment-based study

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Multiple sluicing in Korean

- As in English, Korean also allows *wh*-sluicing, but obligatorily requires the presence of the copula verb *-i-* with interrogative-clause marker *as* *-(nu)nci*.

(1) mimi-ka ecey nwukwunka-lul, manna-ss-nuntey, **nwukwu**-i-nci molu-keyss-ta.
 Mimi-NOM yesterday someone-ACC meet-PST-but who-COP-QUE not.know-PST-DECL
 ‘Mimi met someone yesterday, but I don’t know who.’ (Kim 2015: (6))

- Distinct from English, however, a coordinating conjunction *kuliko* (*and*) optionally occurs with multiple remnants in ellipsis site in Korean.

(2) a. nwukwun-ka hoyuy-lul tten-ass-nuntey, **nwuka encey**-i-nci molu-keyss-ta.
 someone-NOM meeting-ACC leave-PST-but, who when-COP-QUE not.know-PST-DECL
 ‘Someone left the meeting, but I don’t know who when.’
 b. nwukwun-ka hoyuy-lul tten-ass-nuntey, **nwuka (kuliko) encey**-i-nci
 someone-NOM meeting-ACC leave-PST-but, who (and) when-COP-QUE
 molu-keyss-ta.
 not.know-PST-DECL
 ‘Someone left the meeting, but I don’t know who and when.’

Research questions

- To look into the distributional patterns of multiple *wh*-items with or without a coordinating conjunction
- To observe the acceptability between ellipsis and its (contextual) antecedent conducting judgement tasks
- To account for the structural information based on the environmental requirements

Syntactic functions of *wh*-remnant pairs

- Both in English and Korean, remnants could be either an argument or an adjunct, irrelevant of their linear orders.

(3) a. Someone ate something, but I don't remember **who** and **what**.
(argument-argument)

b. Someone ate the pie, but I don't know **who** and **when**. (argument-adjunct)

(4) a. nwukwun-ka mwues-ul mek-ess-nuntey, **nwu-ka** kuliko **mwues**-i-nci
someone-NOM something-ACC eat-PST-but, who-NOM and what-COP-QUE
molu-keyss-ta

not.know-PST-DECL

'Someone ate something, but I don't know who and what.' (argument-argument)

b. nwukwun-ka ppang-ul mek-ess-nuntey, **nwu-ka** **encey**-i-nci
someone-NOM bread-ACC eat-PST-but, who-NOM when-COP-QUE
molu-keyss-ta.

not.know-PST-DECL

'Someone ate the bread, but I don't know who when.' (argument-adjunct)

Syntactic functions of *wh*-remnant pairs (cont'd)

- (5) a. Someone ate the pie, but I don't recall **when** and **who**. (adjunct-argument)
 b. The pie was eaten, but I don't know **by whom** and **when**. (adjunct-adjunct)
- (6) a. nwukwun-ka ppang-ul mek-essn-untey, **encey** kuliko **nwukwu**-i-nci
 someone-NOM bread-ACC eat-PST-but, when and who-COP-QUE
 molu-keyss-ta.
 not.know-PST-DECL
 'Someone ate the bread, but I don't know when and who.' (adjunct-argument)
- b. ppang-ul cwu-ess-nuntey, **nwukw-ueykey** kuliko **encey**-i-nci molu-keyss-ta.
 bread-ACC give-PST-but, who-to and when-COP-QUE not.know-PST-DECL
 '(I) Gave the bread, but I don't know to who(m) and when.' (adjunct-adjunct)

Overttness of the correlate

- Merger and sprouting cases of multiple sluicing can be observed in both English and Korean (Park and Kang 2007, Lasnik 2008, Kim 2013; 2015).

(7) a. Something disappeared, but I don't know **what** and **when**. (Merger)

b. The car left, but I don't remember **when** and **where**. (Sprouting)

(8) a. mwues-i kapang-eyse epsecy-ess-nuntey,

something-NOM bag-from disappear-PST-but,

mwues-i kuliko encey-i-n-ci molu-keyss-ta.

what-NOM and when-COP-QUE not.know-PST-DECL

'Something disappeared from the bag, (but I) don't know what and when.'

(Merger)

b. cha-ka tten-ass-nuntey, **encey eti-eyse-i-n-ci** kkamek-ess-ta.

car-NOM leave-PST-but, when where-from-COP-QUE forget-PST-DECL

'A car left, but I forgot when where.'

(Sprouting)

Coordinating conjunction in multiple *wh*-remnants

- The presence of the coordinating conjunction *and* is obligatory in order for multiple sluicing in English to become grammatical (Citko & Gračanin-Yuksek 2020).

- (9) a. ?* John sang something, but I don't remember **what where**.
 b. John sang something, but I don't remember **what** and **where**.

(Adapted from Citko & Gračanin-Yuksek 2020: (14a))

- As can be seen in (9b), the acceptability of English multiple sluicing improves when conjoined with the coordinating conjunction.
- In Korean multiple sluicing, the coordinating conjunction *kuliko* is optional.

- (10) a. mimi-ka nolay-lul pwull-ess-nuntey, **mwues-ul eti-eyse-i-nci** molu-keyss-ta.
 Mimi-NOM song-ACC sing-PST-but, what-ACC where-from-COP-QUE not.know-PST-DECL
 'Mimi sang a song, but I don't know what where.'
- b. mimi-ka nolay-lul pwull-ess-nuntey,
 Mimi-NOM song-ACC sing-PST-but,
mwues-ul kuliko eti-eyse-i-nci molu-keyss-ta.
 what-ACC and where-from-COP-QUE not.know-PST-DECL
 'Mimi sang a song, but I don't know what and where.'

Case markers in argument remnants

- English multiple sluicing can conjoin subject and object, as well as dative constructions (Lasnik 2007; 2014, Abels and Dayal 2017a).

- (11) a. Someone left, but I don't know **who** and **when**. (Subject)
 b. He met someone, but I forgot **whom** and **where**. (Object)
 c. Someone gave a book to someone, but I don't remember **who** (and) **to whom**. (Dative)

- In contrast, Korean has case markers which restricts the attachment of the copula verb and the interrogative-clause marker after the nominative and accusative markers (Kim 2015).

- (12) nwu-ka khol-ul kol-ass-nuntey,*nwu-ka-i-nci molu-keyss-ta.
 someone-NOM nose-ACC snore-PST-but, *who-NOM-COP-QUE not.know-PST-DECL
 'Someone snored, but I don't know who.'

Two possible readings

- In English, only the Single-Pair (SP) reading is available in coordinated multiple *wh*-sluicing (Whitman 2002, Citko 2013, Citko & Gračanin-Yuksek 2020).

(13) A: Someone left this house, but I don't know **who** and **when**.

B: Mimi left in the afternoon.

(*Single-Pair reading*)

B': *Mimi left in the afternoon and Lee left at midnight.

(*Pair-List reading*)

(Adapted from Citko & Gračanin-Yuksek 2020: 10)

Two possible readings (cont'd)

- In Korean, however, both Single-Pair and Pair-List (PL) readings are available in multiple sluicing (Jung 2018).

(14) A: nwukwun-ka i cip-ul tten-ass-nuntey,

someone-NOM this home-ACC leave-PST-but,

nwuka kuliko **encey**-i-nci molu-keyss-ta.

who and when-COP-QUE not.know-PST-DECL.

‘Someone left this house, but I don’t know who and when.’

B: mimi-ka ohwu-ey tten-ass-ta.

(Single-Pair Reading)

Mimi-NOM afternoon-TEMP leave-PST-DECL

‘Mimi left in the afternoon.’

B’: ? mimi-ka ohwu-ey, minci-ka saypyek-ey tten-ass-ta.

Mimi-NOM afternoonTEMP Minji-NOM midnight-TEMP leave-PST-DECL

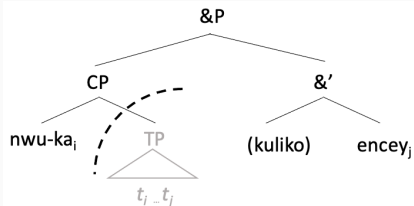
‘Mimi left in the afternoon, Minji left at midnight.’

(Pair-List Reading)

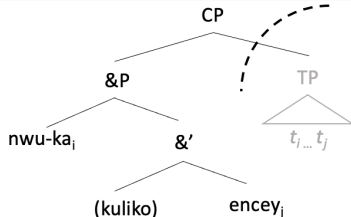
Mono-clausal accounts with a movement-based approach

- Within the Minimalist approach, there are two structures of multiple sluicing: leftward movement with rightward extraposition and multiple *wh*-fronting.
- First remnant moves to SpecCP while the second remnant undergoes rightward extraposition (Lasnik 2008; 2014).
- Multiple *wh*-fronting moves both remnants to SpecCP, but they are not linear because they are in separate domains (Park and Kang 2007, Richards 2010).

(15)



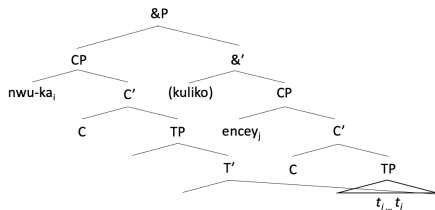
(16)



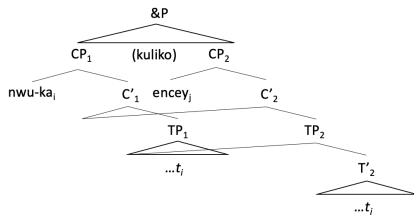
Bi-clausal structure accounts with a movement-based approach

- The bulk-sharing coordinates the remnants at the CP level, sharing the lexical material in the antecedent as a whole (Gračanin-Yuksek 2007, Citko 2013).
- The non-bulk sharing structure does not pose any dominance on the lexical material shared between the remnants (Citko 2013, Citko & Gračanin-Yuksek 2020).

(17)



(18)



Goals of the experiments

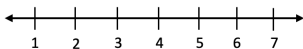
- **Experiment I** investigates the syntactic distributions regarding the grammatical function matching between remnant and its correlate.
- **Experiment II** looks into the semantic aspects of multiple sluicing in Korean with the two possible readings elicited by the use of the construction.
- **Experiment III** examines the restrictions when the copula *-i-* hosts default cases of *wh*-remnants, either with nominative *-i/-ka* or accusative *-ul/-lul* markers, in a multiple *wh*-sluiced question form¹.

Variables

- Independent variables consist of:
 - **Experiment I:**
 - 1) Syntactic matching between grammatical function of remnant and the correlate
 - 2) Existence of coordinating conjunction *kuliko*
 - **Experiment II:**
 - 1) Two possible readings: Single-Pair (SP) or Pair-List (PL) answer types
 - 2) Existence of coordinating conjunction *kuliko*
 - **Experiment III:**
 - 1) The matchedness of case markers between the correlate and the remnant
 - 2) Existence of coordinating conjunction *kuliko* in multiple sluicing in matrix clause
- Dependent variables include:
 - 1) Response time
 - 2) Rating

Participant and procedures

- Participants are targeted with at least **30** native speakers of Korean with the age range of **20-60, without** any cognitive deficiencies for each experiment.
- A 7-point Likert scale will be utilized through PCIBex Farm.



- 96 sentences will be divided into four groups (A, B, C, and D) in order to conduct a Latin square design, with total of 24 (8 target, 16 filler sentences) test sentences in each group.

Trials	Group type	Group type	Group type	Group type
trial 1	A	B	C	D
trial 2	B	C	D	A
trial 3	C	D	A	B
trial 4	D	A	B	C

Experimental design: Fillers and follow-up questions

- There will be 8 stimuli sentences for each type ($4 \times 8 = 32$) and 64 **filler sentences** consisting of related *wh*-constructions.

(19) a. 누가 그리고 언제 이 커피를 마셨니? (Coordinated *wh*-question)

b. 지민이가 밥을 언제 그리고 누구와 먹었는지 궁금하다.

(Embedded coordinated interrogatives)

c. 무언가 얼굴에 생겼는데, 무엇인지 모르겠다. (Sluicing)

- A follow-up question will be given as such to check whether the participants are paying attention to the experiment in the middle of the test process, which will not be applied to the results.

(20) (As a follow-up question to a previous sentence) 창문이 깨졌다.

a. 예.

b. 아니오.

Experiment I: Grammatical functions

- This test will judge the acceptability of CM-slucing in Korean with the following variables:
 - 1) if grammatical function of the *wh*-remnants **match** or **mismatch** with the antecedent
 - 2) whether a conjunction '*kuliko*' coordinating the two *wh*-remnants or no conjunction in between affects the acceptability

(21) MATCHING ARGUMENT-ADJUNCT PAIRS WITH *kuliko*

- a. 내가 분명히 학생에게 자료를 전달했는데, 무엇을 그리고 누구에게인지 까먹었다.

MATCHING ARGUMENT-ADJUNCT PAIRS WITH NO CONJUNCTION

- b. 어제 점심을 먹었는데, 무엇을 언제인지 기억나지 않는다.

MISMATCHING ARGUMENT-ADJUNCT PAIRS WITH *kuliko*

- c. 잠시후에 중간에서 만나기로 약속했는데, 어디를 그리고 언제인지 모른다.

MISMATCHING ARGUMENT-ADJUNCT PAIRS WITH NO CONJUNCTION

- d. 선거에서 시민들이 국회의원을 뽑았는데, 누가 누구로부터인지 궁금하다.

Experiment I: Acceptability predictions

- According to the variables of grammatical function matching and coordinating conjunctions, the four structures are predicted to show varying levels of acceptability.

(22)	MATCHING ARGUMENT-ADJUNCT PAIRS WITH <i>kuliko</i>		More acceptable
	MISMATCHING ARGUMENT-ADJUNCT PAIRS WITH <i>kuliko</i>		
	MATCHING ARGUMENT-ADJUNCT PAIRS WITH NO CONJUNCTION		
	MISMATCHING ARGUMENT-ADJUNCT PAIRS WITH NO CONJUNCTION		Less acceptable

- Our prediction is as follows: regardless of the grammatical function matches between ellipsis-antecedent relations, the existence of a coordinating conjunction *kuliko* seems to enhance the acceptability, leading to a clausal interpretation for each remnant.

Experiment II: Reading preferences

- The second experiment suggest two variables:

1) **Single-Pair** (SP) and **Pair-List** (PL) answers to the multiple sluicing construction

2) a conjunction '*kuliko*' coordinating the two *wh*-remnants or no conjunction in between

(23) Q: 학생이 무엇을 홍보하는데, 누가 그리고 무엇인지 알고 싶다.

A: 지우가 학교 축제를 홍보해. (SP)

A': 지우가 학교 축제를, 미미가 동아리를 홍보해. (PL)

(24) Q: 그들이 아까 집으로 갔는데, 누가 언제인지 모르겠어.

A: 학생들이 오전에 갔대. (SP)

A': 학생들이 오전에, 선생님들이 오후에 갔대. (PL)

Experiment II: Acceptability predictions

- Experiment II consists of four structures with various degrees of acceptability in accordance with two variables: two types of possible readings and a conjunction *kuliko*.

(25) Q: 아까 분명 누가 있었는데 없어졌네. 그런데 언제 그리고 누구인지 난 잘 모르겠어.

A: 지민이가 아침에 잠깐 있었어.

[Single-Pair reading]

A': 지민이가 아침에 잠깐 있었고, 미미가 점심에 있었어.

[Pair-List reading]

(26) Q: 오늘 이미 밥을 먹었다고 하는데, 누가 언제인지 잘 모르겠네.

A: 지우가 열시에 먹었대.

[Single-Pair reading]

A': 지우가 열시에, 미미가 열두시에 먹었대.

[Pair-List reading]

- Our putative expectations: the acceptability of possible readings diverges into SP and PL, determined by the existence of a coordinating conjunction *kuliko*.

Experiment III: Case markers

- Experiment III proposes two variables:

1) the **matchedness** of case markers between the correlate and the remnant

2) when the multiple *wh*-sluiced question is **with** or **without** *kuliko*

(27) a. A: 내가 어제 학생에게 자료를 전달했어.

B: 누구에게 그리고 무엇을?

[matching with *kuliko*]

b. A: 내가 어제 학생에게 자료를 전달했어.

B: 누구에게 무엇을?

[matching with no conjunction]

c. A: 학생이 어제 나에게 자료를 전달했어.

B: 누구를 그리고 무엇이?

[mismatching with *kuliko*]

d. A: 학생이 어제 나에게 자료를 전달했어.

B: 누구를 무엇이?

[mismatching with no conjunction]

- The ordering of the correlates and the remnants is placed in a linear to remove any confusion, which can arise from case mismatching.
- Korean sluicing obligatorily requires the copula verb *-i-* with the following interrogative-clause marker *-(n)ci*, which is not allow to host nominative *-i/-ka* or accusative *-ul/-lul* preceding it (Kim 2013; 2015).

Experiment III: Acceptability predictions

- Four structures can be generated based on the variables of case matchedness and the presence of coordinating conjunctions conjoining the *wh*-remnants, and their respective acceptabilities are predicted to vary.

- (28)
- | | | |
|--|---|-----------------|
| a. CASE MATCHING WITH <i>kuliko</i> |  | More acceptable |
| b. CASE MATCHING NO CONJUNCTION | | |
| c. CASE MISMATCHING WITH <i>kuliko</i> | | |
| d. CASE MISMATCHING NO CONJUNCTION | | Less acceptable |

- Our presumption: since multiple sluicing is in a matrix clause rather than an embedded one, it can be hypothesized that case matching instance with *kuliko* would be the most acceptable, while the case mismatching instance with no conjunction would be the least acceptable.

Experiment I: Grammatical functions

- Grammatical function matching between the *wh*-remnant and the correlate is preferred in terms of acceptability.
- While coordinating conjunction is obligatory in English, *kuliko* in Korean is optional, and therefore does not affect the acceptability of the construction.
- If the *wh*-remnant pair consists of at least one adjunct, it is more preferred than the argument-argument pair.
- When the *wh*-remnant denoting an argument is placed as the second remnant, the sentence is dispreferred.

Experiment II: Reading preferences

- Single Pair (SP) reading is much preferred to the Pair List one.
- In accordance with the first experiment, the presence of *kuliko* does not seem to affect the acceptability of the construction.
- Both argument-adjunct pair and adjunct-argument pair are preferred in Korean, distinct from the acceptability of the latter in English (although it is not verified yet).

Experiment III: Case markers

- The adjunct-argument pair with the accusative case at the second remnant was preferred.
- The cases when *wh*-remnants and their correlates are matched seem to be much more acceptable.
- The conjunction *kuliko* does not affect the acceptability of the construction, which is consistent with the previous two experiments.

Qualitative analysis

- We plan on analyzing the data using R version 4.1.2 (R Core Team 2021) with the *lmerTest Package* (Kunznetsova et al. 2017) for the regression analysis and the mixed effects models.
- Since the data is ordinal, it will be analyzed with cumulative mixed effects models (*clmm*), using the *ordinal* package for R (Christensen 2015).
- We will apply the z-transformation to the ratings, to mend any individual bias from the participants.
- Also we will investigate the distribution of the data in prior to applying the regression analysis.
- The plots will be made for visualization using the *ggplot2* package (Wickham 2016).

Summary

- Multiple sluicing in Korean shows not only similarities but also differences from English in that both allows argument or/and adjunct *wh*-remnant pairs, regardless of the overtness of correlate in the antecedent site.
- However, they differ in the optionality of the coordinating conjunction, especially when the argument remnant restricts the sluicing with certain case markers, which also allows for variability in the two readings.
- As in previous analyses, syntactically distinct structures result in different interpretations that multiple sluicing can have.
- Unlike English, where multiple sluicing is limited to the Single-Pair reading, preferences in the two readings determined by acceptability ratings can account for not only syntactic structure but also semantic behaviors of Korean multiple sluicing.

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